Every finite set of sentences has a tableau in which every branch is complete (fair saturation terminates)

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Editorial note. No step of the counted argument is marked as omitted; the fairness discipline (fixed order, all eligible prefixed formulas treated, propagation through branch tips) is displayed in the delivered span.

Editorial note. The author's concluding sentence reads 'But by construction, every branch is closed'; read against the delivered definition of a complete branch, the construction establishes completeness of every branch (no prefixed formula to which a rule can be applied remains without its conclusion).

Editorial scope. One original result of the Open Logic Project unit normal-modal-logic/tableaux, counted once: existence of a tableau with all branches complete for every finite set. Only the counted statement (source lines 467-469), its own author proof (471-483), and the complete-branch definition above it are delivered.

Allowed standard prerequisites (editorial).

- Prefixed tableau construction. A modal tableau works with prefixed signed formulas S A [sigma]; rules extend branches by appending conclusions, some rules stack conclusions, some branch, some modal rules introduce a new prefix sigma.n, others require a prefix already used on the branch.
- Open and closed branches. A branch is closed if it contains an explicit contradiction; a tableau construction extends open branches; the finite set Gamma supplies the initial assumptions.
- Branch completeness. A branch is complete when the four conditions of the delivered definition hold: stacking conclusions present, a branching conclusion present, at least one new-prefix conclusion present, and used-prefix conclusions present for every used prefix.

Definition 0.1. A branch in tableau is called complete if, whenever it contains a prefixed formula $S(A)[\sigma]$ to which a rule can be applied, it also contains

1. the prefixed formulas that are the corresponding conclusions of the rule, in the case of propositional stacking rules;

2. one of the corresponding conclusion formulas in the case of propositional branching rules;

3. at least one possible conclusion in the case of modal rules that require a new prefix;

4. the corresponding conclusion for every prefix occurring on the branch in the case of modal rules that require a used prefix.

Proposition 0.2. *Every finite* Γ *has tableau in which every branch is complete.*

Proof. Consider an open branch in tableau for Γ. There are finitely many prefixed formulas in the branch to which a rule could be applied. In some fixed order (say, top to bottom), for each of these prefixed formulas for which the conditions (1)–(4) do not already hold, apply the rules that can be applied to it to extend the branch. In some cases this will result in branching; apply the rule at the tip of each resulting branch for all remaining prefixed formulas. Since the number of prefixed formulas is finite, and the number of used prefixes on the branch is finite, this procedure eventually results in (possibly many) branches extending the original branch. Apply the procedure to each, and repeat. But by construction, every branch is closed. □